S&P GLOBAL & CRISIL CAMPUS HACKATHON

Financial Risk Intelligence Engine

Turning market news into explainable portfolio stress signals

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THE PROBLEM

Signals over headlines

Risk teams need a consistent input for action.

Risk teams need signals, not more headlines

01 **Unstructured in**

Market news and commentary arrive as unstructured text, but downstream risk workflows need consistent inputs.

02 **Structured through**

Our approach automates ingestion, entity extraction, FinBERT sentiment, event classification, explainable impact scoring, portfolio mapping, and stress testing.

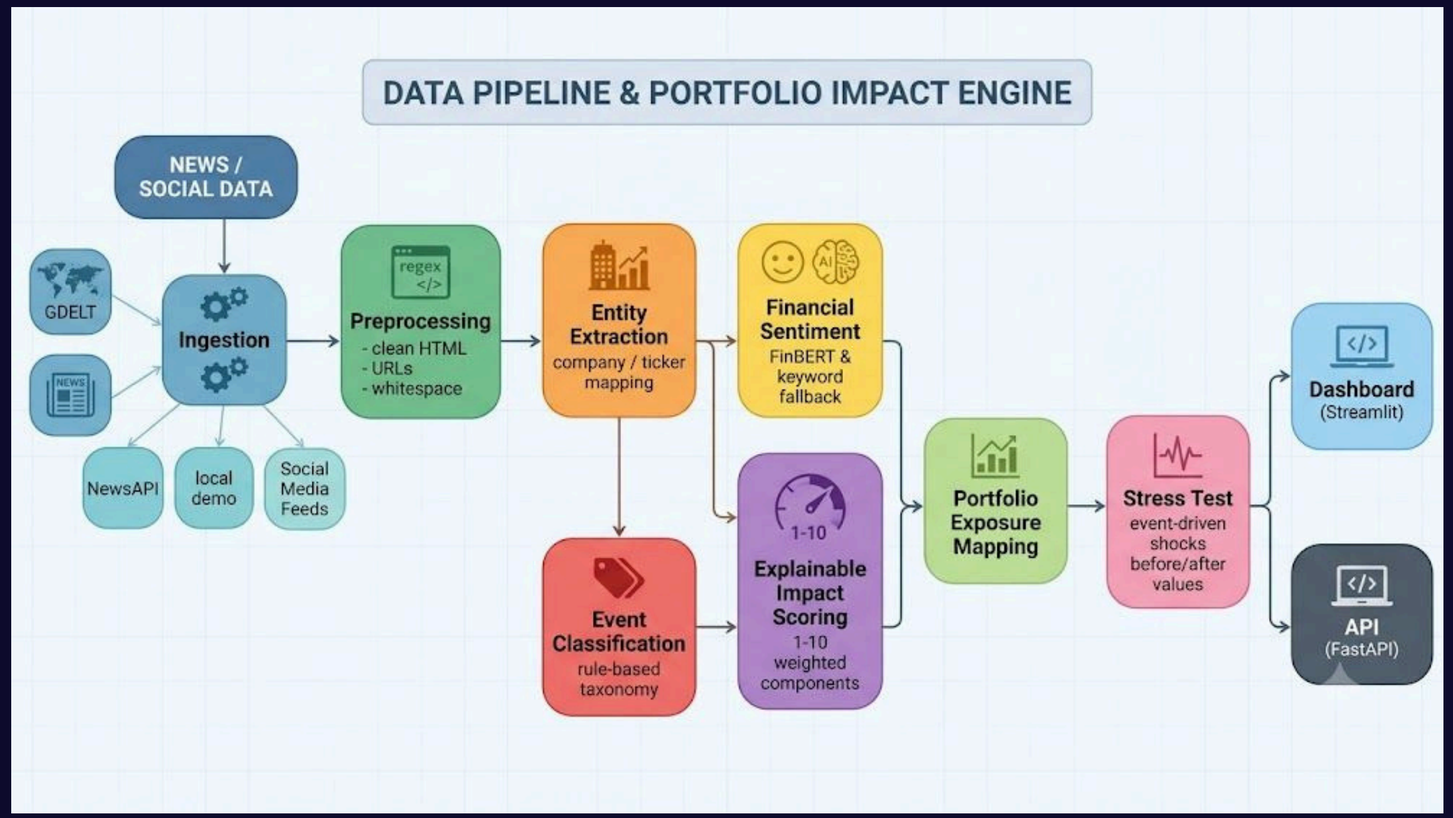
03 **Actionable out**

Output: a structured risk signal that can trigger action.

SYSTEM ARCHITECTURE

From news to a stress test in one pipeline

A high-impact signal launches the stress test.



High-impact signals branch into stress testing before dashboard review.

IMPLEMENTATION

Built for speed, transparency, and local-first demos

A modular stack keeps the prototype inspectable from endpoint to stress output.

FastAPI

REST backend with analyze, portfolio, stress-test, risk-signals, and demo endpoints.

Streamlit

Interactive dashboard for signal and portfolio impact review.

FinBERT

Financial sentiment with a keyword fallback when the model is unavailable.

Rule-based taxonomy

Transparent event classification and explainable heuristics.

Synthetic portfolio

Illustrative $100M stress engine.

PROTOTYPE EVIDENCE

Prototype proves the full feedback loop

FINBERT ACCURACY

0.889

on the 4,845-headline Financial PhraseBank evaluation set

TRIGGER

≥7

Signals with impact score ⩾7 automatically trigger a stress test with before/after portfolio value, loss, and per-asset impact.

End-to-end extraction turns financial news into entity, sentiment, event, and impact signals on a 1–10 scale.



OPERATING VALUE

Noise becomes an actionable risk workflow

Concise by design

Risk teams get a concise, explainable signal instead of scanning every article manually.

Modular by design

Modular stages let teams add data sources, refine entity coverage, or swap in calibrated shock tables without rewriting the core pipeline.

Useful under constraint

Demonstrates how AI/NLP can support portfolio stress testing, scenario analysis, and high-impact event monitoring under hackathon constraints.



SCALE PATH

Next step: move from heuristics to calibration

01

Current limits

Heuristic impact scoring, restricted entity map, and illustrative synthetic shocks.

02

Near-term

Better article body extraction, persistent signal history, and richer company/entity coverage.

03

Scale path

Calibrated shock tables plus sector- and geography-specific stress mappings.

Convert unstructured market information into a repeatable risk decision loop.